

第十周作业参考解答

p. 155, 习题 14.2, 1, 2(2), (3), 3, 4, 5, 6, 7.

1. 解: (1) 线段 AB 的参数方程为 $\begin{cases} x = t+1, \\ y = 2t+1, \end{cases} \quad t: 0 \rightarrow 1$, 则

$$\int_{L^+} xy dx + (y-x) dy = \int_0^1 (2t^2 + 5t + 1) dt = \frac{25}{6}.$$

$$(2) \quad dy = 4(x-1)dx, \text{ 则 } \int_{L^+} xy dx + (y-x) dy = \int_1^2 (10x^3 - 32x^2 + 35x - 12) dx = \frac{10}{3}.$$

(3) 线段 $\overline{AD}: y=1, x: 1 \rightarrow 2$; 线段 $\overline{DB}: x=2, y: 1 \rightarrow 3$, 则

$$\begin{aligned} \int_{L^+} xy dx + (y-x) dy &= \int_{\overline{AD}} xy dx + (y-x) dy + \int_{\overline{DB}} xy dx + (y-x) dy \\ &= \int_1^2 x dx + \int_1^3 (y-2) dy = \frac{3}{2}. \end{aligned}$$

2. 解: (2) 曲线 L^+ 的参数方程 $x = a \cos \theta, y = a \sin \theta, \theta: 0 \rightarrow 2\pi$, 则

$$\begin{aligned} &\int_{L^+} \frac{(x+y)dx + (y-x)dy}{x^2 + y^2} \\ &= \int_0^{2\pi} \frac{a^2(\cos t + \sin t)(-\sin t) + a^2(\sin t - \cos t)\cos t}{a^2} dt \\ &= -\int_0^{2\pi} dt = -2\pi. \end{aligned}$$

(3) $dx = a(1 - \cos t)dt, dy = a \sin t dt$, 则

$$\begin{aligned} \int_{L^+} \frac{x}{y} dx + \frac{1}{y-a} dy &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (a(t - \sin t) - \tan t) dt \\ &= \frac{at^2}{2} + a \cos t + \ln |\cos t| \Big|_{\frac{\pi}{6}}^{\frac{\pi}{3}} = \frac{\pi^2 a}{24} + \frac{1-\sqrt{3}}{2} a - \frac{\ln 3}{2}. \end{aligned}$$

3. (1) 线段 $\overline{AB}$ 的参数方程为 $x=1+t, y=1+2t, z=1+3t, t: 0 \rightarrow 1$, 因此

$$\int_{\overline{AB}} x dx + y dy + z dz = \int_0^1 (6+14t) dt = 13.$$

(2)

$$\int_{L^+} \frac{xdy - ydx}{x^2 + y^2} + b dz = \int_0^{2\pi} \left(\frac{a^2 \cos^2 t + a^2 \sin^2 t}{a^2} + b^2 \right) dt = \int_0^{2\pi} (1+b^2) dt = 2\pi(1+b^2).$$

4. 解: 曲线 L^+ 在 Oxy 坐标平面上的投影曲线是 L_1^+ : $\begin{cases} x^2 + 2y^2 = 1 \\ z = 0, \end{cases}$ 逆时针方向为正方向,

其参数方程为: $x = \cos t, y = \frac{\sqrt{2}}{2} \sin t$, 参数 t 从 $t = 0$ 到 $t = 2\pi$, 由于曲线 L 在平面 $z = y$ 上,

$$\text{因此 } \int_{L^+} xyz dz = \frac{\sqrt{2}}{4} \int_0^{2\pi} \cos^2 t \sin^2 t dt = \frac{\sqrt{2}}{16} \pi.$$

5. 解法一: 我们考虑将第二型曲线积分转化为第一型曲线积分计算。先求 L^+ 的单位切向量 τ , 由于 L^+ 在平面 $x + y + z = 0$ 上, 因此 τ 垂直于该平面的法向量 $\mathbf{n}_1 = (1, 1, 1)$, 又 L^+ 在球面 $x^2 + y^2 + z^2 = 1$ 上, 因此 τ 垂直于该球面的法向量 $\mathbf{n}_2 = (x, y, z)$, 所以 τ 平行于 $\mathbf{n}_1 \times \mathbf{n}_2 = (z - y, x - z, y - x)$. 由于 $\mathbf{n}_1, \mathbf{n}_2, \tau$ 构成右手系, 故

$$\tau = \frac{\mathbf{n}_1 \times \mathbf{n}_2}{\|\mathbf{n}_1 \times \mathbf{n}_2\|} = \frac{(z - y, x - z, y - x)}{\sqrt{(z - y)^2 + (x - z)^2 + (y - x)^2}},$$

从而

$$\begin{aligned} I &= \int_{L^+} z dx + x dy + y dz \\ &= \int_L (z, x, y) \cdot \tau dl \\ &= \int_L \frac{z(z - y) + x(x - z) + y(y - x)}{\sqrt{(y - z)^2 + (z - x)^2 + (x - y)^2}} dl \\ &= \int_L \frac{\sqrt{3}}{2} ds = \sqrt{3}\pi. \end{aligned}$$

解法二、曲线 $L: \begin{cases} x^2 + y^2 + z^2 = 1 \\ x + y + z = 0 \end{cases}$ 在 Oxy 坐标平面上的投影曲线为 $L_1: \begin{cases} x^2 + y^2 + yx = \frac{1}{2} \\ z = 0. \end{cases}$

由曲线 L^+ 的方向可知平面曲线 L_1^+ 是逆时针方向, 其参数方程

$$x = \frac{1}{\sqrt{2}} \cos \theta - \frac{1}{\sqrt{6}} \sin \theta, \quad y = \frac{2}{\sqrt{6}} \sin \theta, \quad \theta: 0 \rightarrow 2\pi.$$

因为曲线 L 位于平面 $z = -x - y$ 上, 因此

$$\begin{aligned}
 I &= \int_{L^+} zdx + xdy + ydz \\
 &= \int_{L_1^+} (-x-y)dx + xdy + y(-dx-dy) \\
 &= \int_{L_1^+} (-x-2y)dx + (x-y)dy = \int_0^{2\pi} \frac{\sqrt{3}}{2} d\theta = \sqrt{3}\pi.
 \end{aligned}$$

6. 解: 曲线 $L^+ : \begin{cases} x^2 + y^2 = 1 \\ z = ax + by \end{cases}$ 投影在 Oxy 坐标平面上是曲线 $L_1 : x^2 + y^2 = 1$, 其正向是逆时针方向, 则沿空间曲线 L^+ 的曲线积分转化为沿平面曲线 L_1^+ 上的曲线积分:

$$\int_{L^+} (y-z)dx + (z-x)dy + (x-y)dz = \int_{L_1^+} (1-a-b)ydx + (a+b-1)xdy.$$

曲线 $L_1^+ : \begin{cases} x = \cos \theta \\ y = \sin \theta, \end{cases} \theta : 0 \rightarrow 2\pi$, 则

$$\int_{L_1^+} (1-a-b)ydx + (a+b-1)xdy = (a+b-1) \int_0^{2\pi} (\sin^2 \theta + \cos^2 \theta) d\theta = 2(a+b-1)\pi.$$

7. 曲线 L^+ 按照正方向依次由下述三段曲线构成:

$L_1^+ : x = \cos t, y = \sin t, z = 0, t \in [0, \frac{\pi}{2}]$, 正方向从 $t = 0$ 到 $t = \frac{\pi}{2}$;

$L_2^+ : x = 0, y = \cos t, z = \sin t, t \in [0, \frac{\pi}{2}]$, 正方向从 $t = 0$ 到 $t = \frac{\pi}{2}$;

$L_3^+ : x = \cos t, y = 0, z = \sin t, t \in [0, \frac{\pi}{2}]$, 正方向从 $t = \frac{\pi}{2}$ 到 $t = 0$.

于是

$$\int_{L_1^+} (y^2 - z^2)dx + (z^2 - x^2)dy + (x^2 - y^2)dz = \int_{L_1^+} y^2 dx - x^2 dy = \int_0^{\frac{\pi}{2}} (-\sin^3 t - \cos^3 t) dt = -\frac{4}{3}.$$

完全对称地, 可以得到

$$\int_{L_2^+} (y^2 - z^2)dx + (z^2 - x^2)dy + (x^2 - y^2)dz = \int_{L_3^+} (y^2 - z^2)dx + (z^2 - x^2)dy + (x^2 - y^2)dz = -\frac{4}{3}.$$

$$\text{因此 } \int_{L^+} (y^2 - z^2)dx + (z^2 - x^2)dy + (x^2 - y^2)dz = -\frac{4}{3} - \frac{4}{3} - \frac{4}{3} = -4.$$

p. 166, 习题 14.3, 1, 2, 3, 5, 7, 8, 13, 14(1), (3), 15.

1. 解: (1) 因为 $P(x, y) = \frac{(x+y)}{x^2+y^2} \in C^1(D)$, $Q(x, y) = \frac{(y-x)}{x^2+y^2} \in C^1(D)$, 且 $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$,

因此由格林公式, $\int_{L^+} \frac{(x+y)dx+(y-x)dy}{x^2+y^2} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = 0.$

$$(2) \int_{L^+} \frac{(x+y)dx+(y-x)dy}{x^2+y^2} = \frac{1}{a^2} \oint_{L^+} (x+y)dx + (y-x)dy = \frac{1}{a^2} \iint_D (-2) dx dy = -2\pi.$$

2. 解: (1) 由格林公式,

$$\int_{L^+} \ln \frac{2+y}{1+x^2} dx + \frac{x(y+1)}{2+y} dy = \iint_D \frac{y+1-1}{y+2} dx dy = \int_{-1}^1 dx \int_{-1}^1 \frac{y}{y+2} dy = 4(1-\ln 3).$$

(2) 补一条从 $(1,0)$ 到 $(\pi+1,0)$ 的线段 l , 则由格林公式,

$$\begin{aligned} & \int_{L^+} \sqrt{x^2+y^2} dx + y \left[xy + \ln(x+\sqrt{x^2+y^2}) \right] dy \\ &= \int_{(L \cup l)^+} (\sqrt{x^2+y^2} dx + y \left[xy + \ln(x+\sqrt{x^2+y^2}) \right] dy) - \int_1^{\pi+1} x dx \\ &= \iint_D y^2 dx dy - \int_1^{\pi+1} x dx = \int_1^{\pi+1} \left(\frac{1}{3} \sin^3(x-1) - x \right) dx = \frac{4}{9} - \frac{\pi^2+2\pi}{2}. \end{aligned}$$

(3) 记 $\left(\frac{\pi}{2}, 1\right)$ 到 $\left(\frac{\pi}{2}, 0\right)$ 的有向线段为 l_1^+ , 记 $\left(\frac{\pi}{2}, 0\right)$ 到 $(0,0)$ 的有向线段为 l_2^+ ,

令 $P(x,y) = 2xy^3 - y^2 \cos x$, $Q(x,y) = 1 - 2y \sin x + 3x^2 y^2$, 则 $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0$. 故由格林

公式,

$$\begin{aligned} & \int_{L^+} (2xy^3 - y^2 \cos x) dx + (1 - 2y \sin x + 3x^2 y^2) dy \\ &= \int_{(L \cup l_1 \cup l_2)^+} P dx + Q dy - \int_{l_1^+} P dx + Q dy - \int_{l_2^+} P dx + Q dy \\ &= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy - \int_1^0 (1 - 2y + \frac{3\pi^2}{4} y^2) dy \\ &= \frac{\pi^2}{4}. \end{aligned}$$

3. 解: 设 $P_0(x_0, y_0)$, 则 $\mathbf{v} = (x-x_0, y-y_0)$, 且 $v = \sqrt{(x-x_0)^2 + (y-y_0)^2}$, 故

$$\begin{aligned} & \int_L \frac{\cos \langle \mathbf{n}, \mathbf{v} \rangle}{v} dl = \int_L \frac{\mathbf{v} \cdot \mathbf{n}}{v^2} dl \\ &= \int_L \left(\frac{x-x_0}{(x-x_0)^2 + (y-y_0)^2}, \frac{y-y_0}{(x-x_0)^2 + (y-y_0)^2} \right) \cdot \mathbf{n} dl \\ &= \int_L \frac{x-x_0}{(x-x_0)^2 + (y-y_0)^2} dy - \frac{y-y_0}{(x-x_0)^2 + (y-y_0)^2} dx, \end{aligned}$$

若 $P_0 \notin D$, 则由格林公式可得原式 $= 0$,

若 $P_0 \in D$, 则取足够小的正数 ε 使圆周 $(x-x_0)^2 + (y-y_0)^2 = \varepsilon^2$ 落在区域 D 内, 记圆周为 l , 逆时针为其正向, L 与 l 围成的区域为 Ω , 则

$$\begin{aligned} \int_{L^+} \frac{\cos \langle \mathbf{n}, \mathbf{v} \rangle}{v} dl &= \int_{(\partial\Omega)^+} \frac{\cos \langle \mathbf{n}, \mathbf{v} \rangle}{v} dl + \int_{l^+} \frac{\cos \langle \mathbf{n}, \mathbf{v} \rangle}{v} dl \\ &= \int_{l^+} \left(\frac{x-x_0}{(x-x_0)^2 + (y-y_0)^2}, \frac{y-y_0}{(x-x_0)^2 + (y-y_0)^2} \right) \cdot \mathbf{n} dl \\ &= \frac{1}{\varepsilon^2} \int_{l^+} (x-x_0) dy - (y-y_0) dx \\ &= \frac{2}{\varepsilon^2} \iint_{(x-x_0)^2 + (y-y_0)^2 \leq \varepsilon^2} dx dy = 2\pi. \end{aligned}$$

证毕

5. (1) 解: 参数方程为 $\begin{cases} x = b \cos \theta \\ y = a \sin \theta \end{cases}, \theta: 0 \rightarrow \arctan \frac{b}{a}$, 则两椭圆在第一象限内所包围的区域面积

$$S_1 = \int_{L^+} x dy - y dx = \int_0^{\arctan \frac{b}{a}} ab(\sin^2 \theta + \cos^2 \theta) d\theta = ab \arctan \frac{b}{a},$$

故两椭圆所包围平面区域的面积 $S = 4S_1 = 4ab \arctan \frac{b}{a}$.

- (2) 令 $x = \rho \cos \theta, y = \rho \sin \theta$, 由方程 $(x^2 + y^2)^2 = a^2(x^2 - y^2)$ 可知双纽线在 y 轴右侧的曲线方程为 $\rho^2 = a^2 \cos 2\theta, \theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$, 从而 $x = a \cos \theta \sqrt{\cos 2\theta}, y = a \sin \theta \sqrt{\cos 2\theta}$,

$$\text{故 } S = \int_{L^+} x dy - y dx = a^2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos 2\theta d\theta = a^2.$$

- (3) 设 L^+ 为摆线与 x 轴围成图形的正向边界, 则该图形的面积

$$\begin{aligned} S &= \frac{1}{2} \int_{2\pi}^0 [a^2(t - \sin t) \sin t - a^2(1 - \cos t)^2] dt + \int_0^{2\pi a} -0 dx \\ &= -\frac{1}{2} \int_0^{2\pi} a^2(t \sin t + 2 \cos t - 2) dt = 3\pi a^2. \end{aligned}$$

7. 解:

$$\oint_{L^+} \frac{\partial u}{\partial \mathbf{n}} dl = \oint_{L^+} (u'_x, u'_y) \cdot (dy, -dx)$$

$$= \oint_{L^+} u'_x dy - u'_y dx = \iint_{x^2+y^2 \leq 6x} (u''_{xx} + u''_{yy}) dx dy = 36\pi.$$

8. 证明: (1)

$$\oint_{\partial D} v \frac{\partial u}{\partial \mathbf{n}} dl = \oint_{\partial D} -\frac{\partial u}{\partial y} v dx + \frac{\partial u}{\partial x} v dy$$

$$= \iint_D (v \frac{\partial^2 u}{\partial y^2} + \frac{\partial v}{\partial y} \frac{\partial u}{\partial y} + v \frac{\partial^2 u}{\partial x^2} + \frac{\partial v}{\partial x} \frac{\partial u}{\partial x}) dx dy$$

$$= \iint_D v \Delta u dx dy + \iint_D \nabla v \cdot \nabla u dx dy.$$

(2) 由(1)知

$$\oint_{\partial D} v \frac{\partial u}{\partial \mathbf{n}} dl = \iint_D v \Delta u dx dy + \iint_D \nabla v \cdot \nabla u dx dy,$$

$$\oint_{\partial D} u \frac{\partial v}{\partial \mathbf{n}} dl = \iint_D u \Delta v dx dy + \iint_D \nabla v \cdot \nabla u dx dy,$$

所以,

$$\oint_{\partial D} \begin{vmatrix} \frac{\partial u}{\partial \mathbf{n}} & \frac{\partial v}{\partial \mathbf{n}} \\ u & v \end{vmatrix} dl = \oint_{\partial D} \left(v \frac{\partial u}{\partial \mathbf{n}} - u \frac{\partial v}{\partial \mathbf{n}} \right) dl$$

$$= \iint_D v \Delta u dx dy + \iint_D \nabla v \cdot \nabla u dx dy - \iint_D u \Delta v dx dy - \iint_D \nabla v \cdot \nabla u dx dy$$

$$= \iint_D \begin{vmatrix} \Delta u & \Delta v \\ u & v \end{vmatrix} dx dy.$$

13. 解: (1) 因为 $\frac{\partial(f(xy)y)}{\partial y} = f'(xy)xy + f(xy)$, $\frac{\partial(f(xy)y)}{\partial x} = f'(xy)xy + f(xy)$,

因此 $\frac{\partial(f(xy)y)}{\partial x} = \frac{\partial(f(xy)y)}{\partial y}$, 所以曲线积分与积分路径无关, 故

$$\oint_{L^+} f(xy)(ydx + xdy) = 0. \text{ 证毕}$$

(2) 因为 $\frac{\partial(f(x^2+y^2)x)}{\partial y} = f'(x^2+y^2)2xy$, $\frac{\partial(f(x^2+y^2)y)}{\partial x} = f'(x^2+y^2)2xy$,

因此 $\frac{\partial(f(x^2+y^2)y)}{\partial x} = \frac{\partial(f(x^2+y^2)x)}{\partial y}$, 所以曲线积分与积分路径无关, 故

$$\oint_L f(x^2+y^2)(xdx+ydy) = 0. \text{ 证毕}$$

14. 解: (1) 将微分式分项组合,

$$\begin{aligned} du &= \left(1 - \frac{1}{y} + \frac{y}{z}\right)dx + \left(\frac{x}{z} + \frac{x}{y^2}\right)dy - \frac{xy}{z^2}dz \\ &= dx + \left(\frac{x}{y^2}dy - \frac{1}{y}dx\right) + \left(\frac{y}{z}dx + \frac{x}{z}dy - \frac{xy}{z^2}dz\right) \\ &= dx - d\frac{x}{y} + d\frac{xy}{z} = d\left(x - \frac{x}{y} + \frac{xy}{z}\right), \end{aligned}$$

所以原函数 $u(x, y, z) = x - \frac{x}{y} + \frac{xy}{z} + C$.

(3) 令 $P = x^2 + 2xy - y^2$, $Q = x^2 - 2xy - y^2$. 由于 $\frac{\partial P}{\partial y} = 2x - 2y = \frac{\partial Q}{\partial x}$, 所以曲线积分

与积分路径无关, 故原函数

$$\begin{aligned} u(x, y) &= \int_{(0,0)}^{(x,y)} P(v, w)dv + Q(v, w)dw + C \\ &= \int_0^x P(v, 0)dv + \int_0^y Q(x, w)dw + C \\ &= \int_0^x v^2 dv + \int_0^y (x^2 - 2xw - w^2)dw + C \\ &= \frac{x^3}{3} + x^2y - xy^2 - \frac{1}{3}y^3 + C. \end{aligned}$$

15. 解: 由于曲线积分 $\int_L xy^2 dx + y\varphi(x)dy$ 与积分路径无关, 因此 $y\varphi'(x) = 2xy$, 从而

$\varphi'(x) = 2x$ 且 $\varphi(x) = x^2 + C$. 由于 $\varphi(0) = 0$, 因此 $\varphi(x) = x^2$ 且

$$\begin{aligned} I &= \int_{(0,0)}^{(1,0)} xy^2 dx + y\varphi(x)dy + \int_{(1,0)}^{(1,1)} xy^2 dx + y\varphi(x)dy \\ &= \int_0^1 y\varphi(1)dy = \int_0^1 ydy = \frac{1}{2}. \end{aligned}$$